ME3500J

Design Review

Group 12

Meng Peilin, Liu Xingjian, Zhu Yufei, Wei Yushan

01.

Introduction



Background

01 **Widely used in infrastructure**

02 **Difficult-to-access environments**

03 **Risky and time-consuming inspections**



A small, automatic robot capable of adhering to vertical cylindrical surfaces is needed.

Literature Review - 1

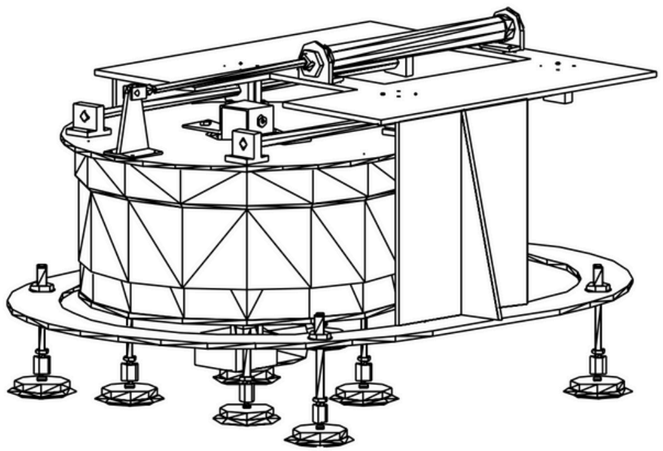


Fig. 1. The robot SURFY.

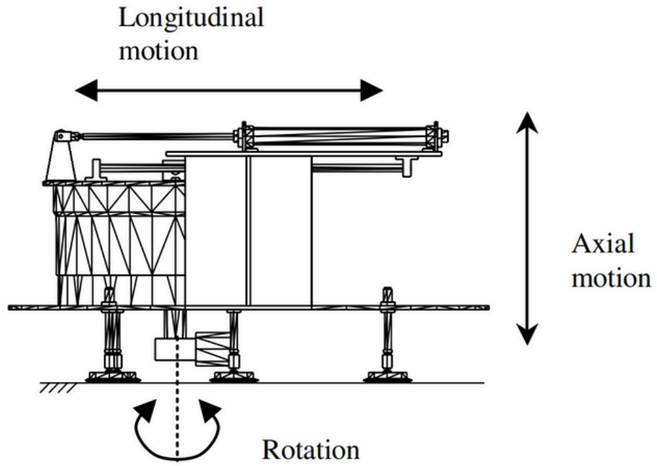


Fig. 2. Description of movements.

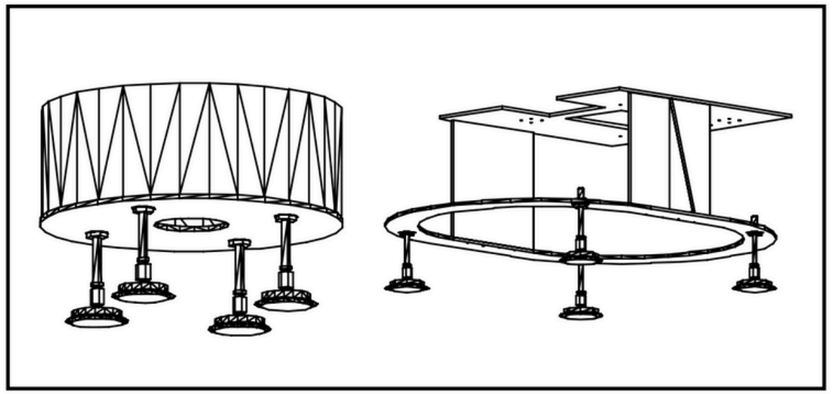


Fig. 3. Internal and external bodies with suction cups.

Motion Structure

- Longitudinal & axial cylinder
- Guide rails
- External & internal frame with suction cups

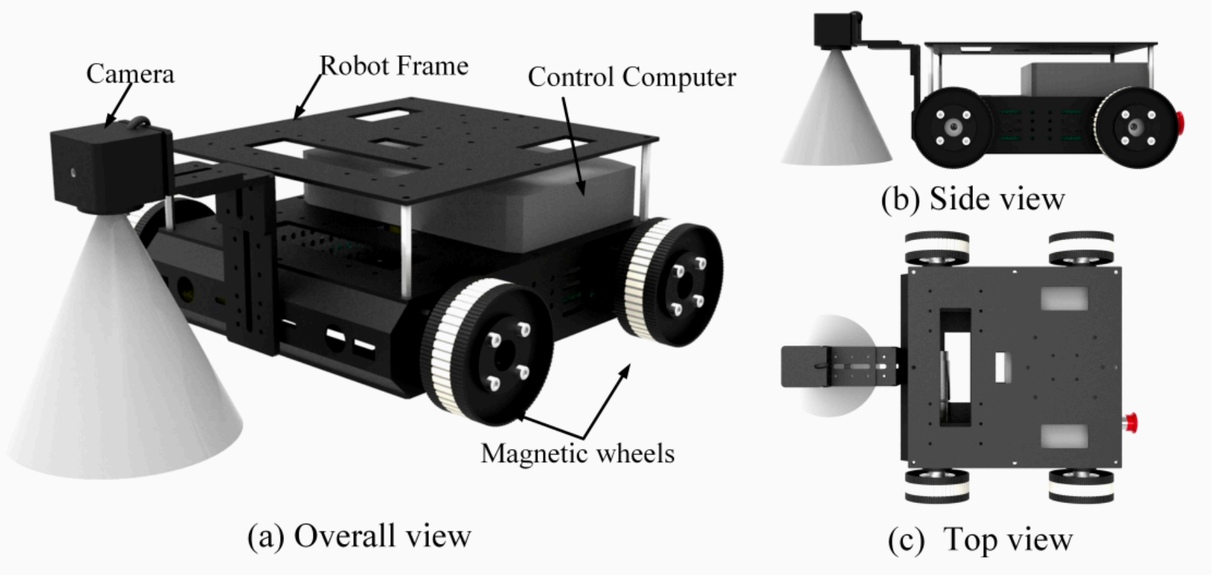
Locking System

- Vacuum pump → negative pressure
- Solenoid valves → suction cups' adhesion and release

Limitations

- Surface smoothness
- Maintenance

Literature Review - 2



Motion Structure

Synchronous belt:

one DC motor for two magnetic wheels

Locking System

Magnetic wheels: annular permanent magnets
→ strong magnetic field
→ adhere to metal surface

Limitations

- **Surface material restriction**
- **Obstacle limitation**

02.
Objectives



Key Points

01 **Fail-safe Locking**

02 **Bi-directional Control**

03 **Position Accuracy**

04 **Time Limits**

- **Phase 1: Ascent & Zero-Power Lock (Ø 0.3m Pier)**
  - Robot ascends vertically to **1.2m height** on a **0.3m-diameter pier**.
  - Upon reaching 1.2m, **power is cut** (remote).
  - The robot must **remain locked in place without power for 60 seconds** (no slipping >1 cm).
  - After 60s, power is restored; robot descends to base autonomously.
- **Phase 2: Transition & Second Ascent (Ø 0.6m Pier)**
  - The robot is repositioned at the base of a **0.6 m-diameter pier**.
  - Ascends to **1.2m height**, power cut again, and maintains position **without power for 60 seconds**.
  - Power restored; descends to base.

03.

Project Plan



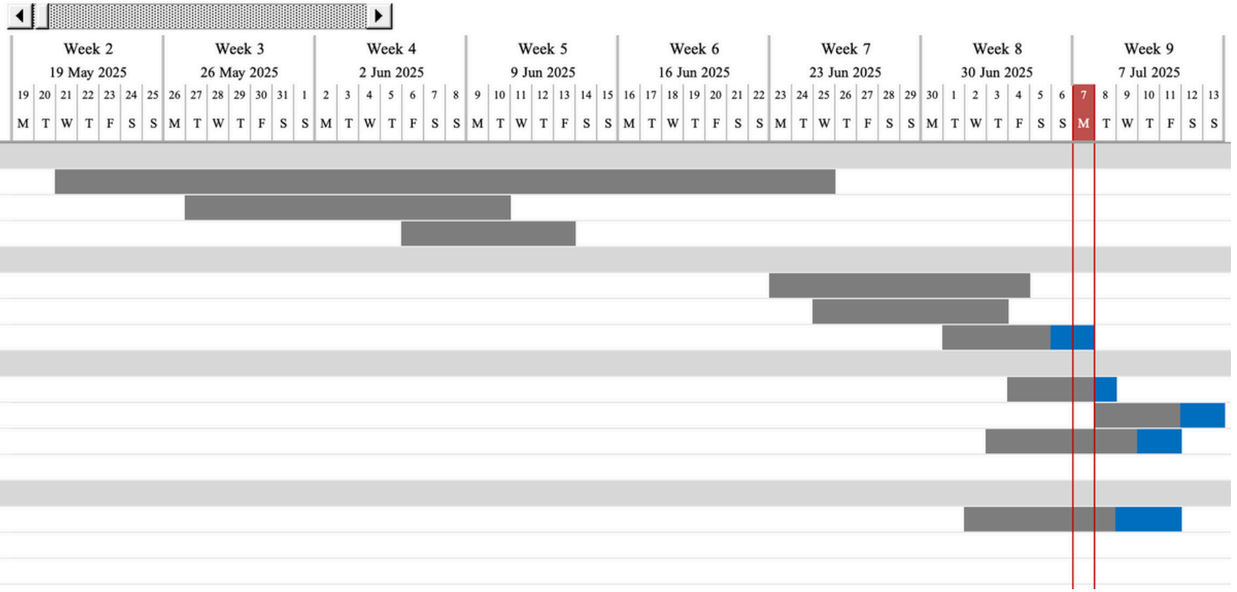
Gantt Chart

[Mechatronic Climbing Robot for Small-Scale Cylindrical Pier Inspection] Project

[Group 12]

Project Start Date: 5/12/2025 (Monday)
Project Lead: Group 12
Display Week: 2

WBS	TASK	LEAD	START	END	DAYS	% DONE	WORK DAYS
1	**[Design]**			-			-
1.1	Brainstorm	All group members	Wed 5/21/25	Wed 6/25/25	36	100%	26
1.2	Lead Screw Mechanism	Liu Xingjian, Meng Peilin	Tue 5/27/25	Tue 6/10/25	15	100%	11
1.3	Soft Gripper	Wei Yushan, Zhu Yufei	Fri 6/06/25	Fri 6/13/25	8	100%	6
2	**[Prototype]**			-			-
2.1	Lead Screw Mechanism	Liu Xingjian, Meng Peilin	Mon 6/23/25	Fri 7/04/25	12	100%	10
2.2	Soft Gripper (gripper only)	Wei Yushan, Zhu Yufei	Wed 6/25/25	Thu 7/03/25	9	100%	7
2.3	Test	All group members	Tue 7/01/25	Mon 7/07/25	7	80%	5
3	**[Manufacturing]**			-			-
3.1	Optimized CAD	Meng Peilin	Fri 7/04/25	Tue 7/08/25	5	80%	3
3.2	Assembly	Wei Yushan, Zhu Yufei	Tue 7/08/25	Wed 7/16/25	9	50%	7
3.3	Programming	Liu Xingjian	Thu 7/03/25	Fri 7/11/25	9	80%	7
3.4	Test	All group members	Wed 7/16/25	Tue 7/22/25	7	0%	5
4	**[Report and Presentation]**			-			-
4.1	Design Review	All group members	Wed 7/02/25	Fri 7/11/25	10	70%	8
4.2	Final Presentation	All group members	Fri 7/18/25	Fri 7/25/25	8	0%	6
4.3	Reprot	All group members	Wed 7/23/25	Tue 7/29/25	7	0%	5

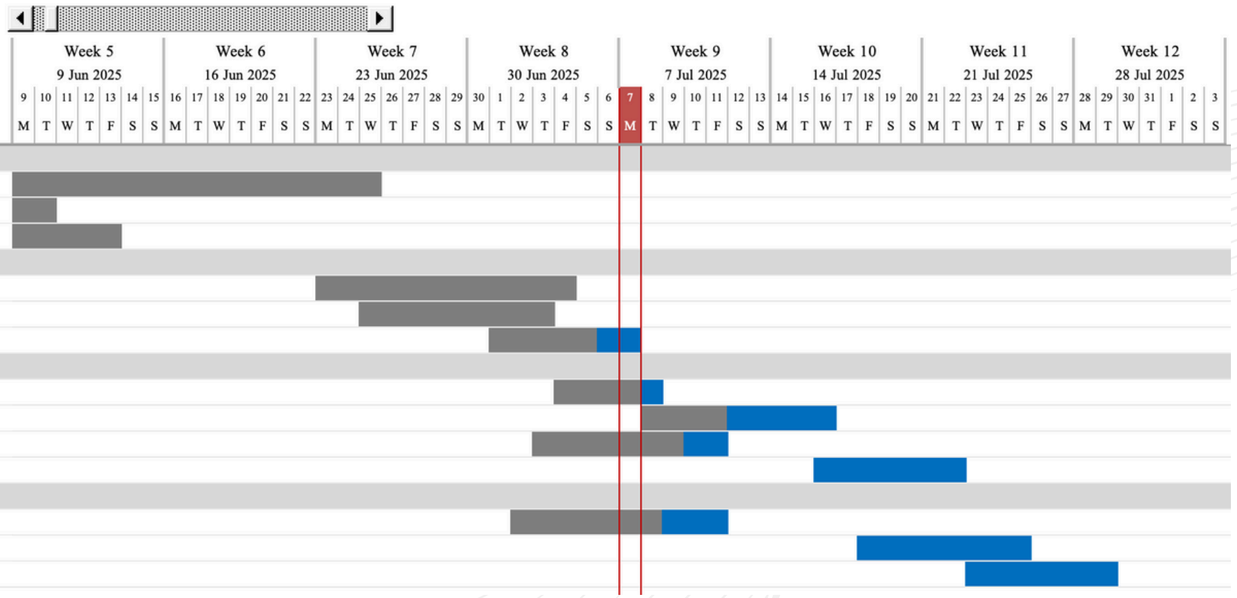


[Mechatronic Climbing Robot for Small-Scale Cylindrical Pier Inspection] Project

[Group 12]

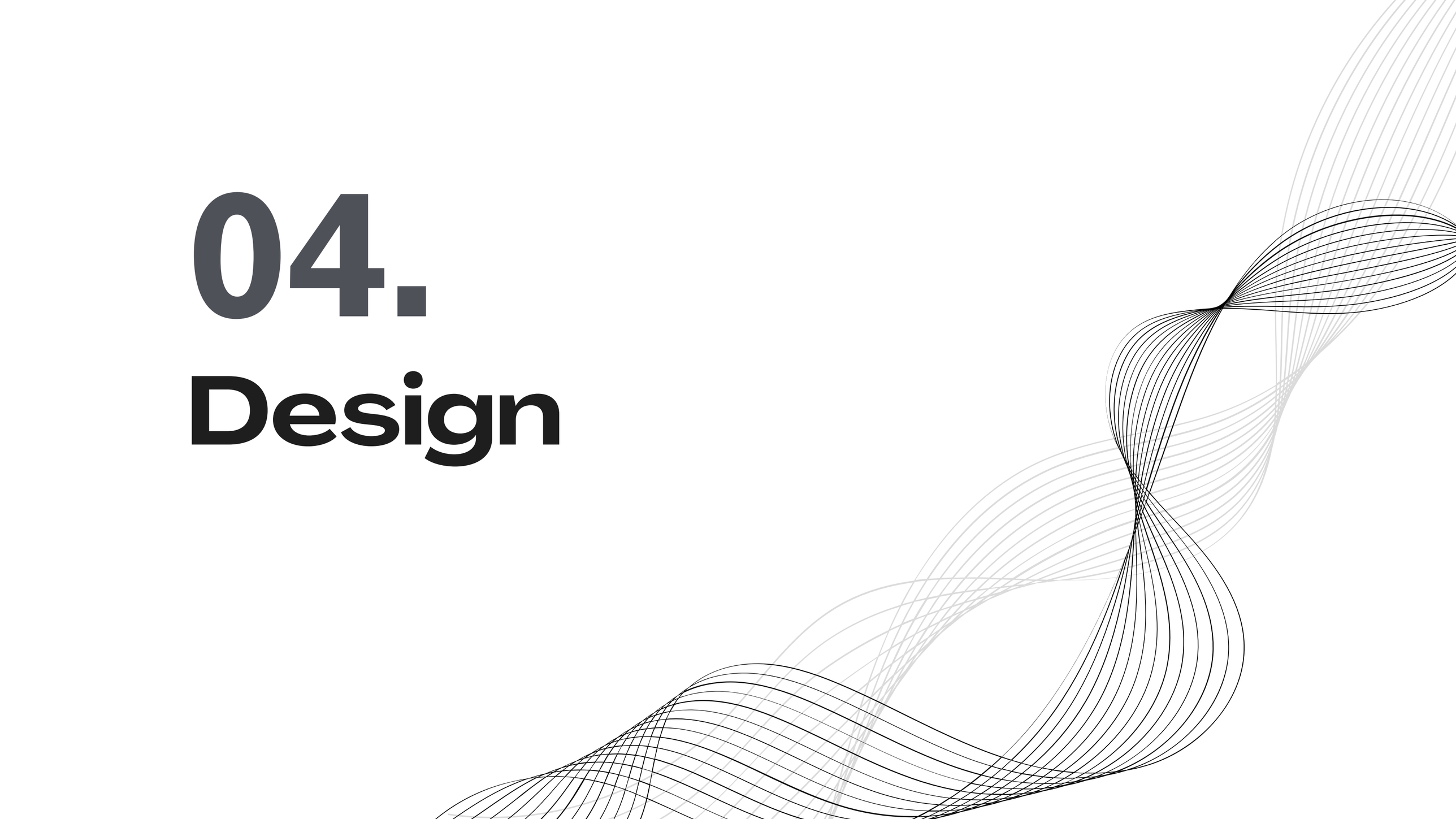
Project Start Date: 5/12/2025 (Monday)
Project Lead: Group 12
Display Week: 5

WBS	TASK	LEAD	START	END	DAYS	% DONE	WORK DAYS
1	**[Design]**			-			-
1.1	Brainstorm	All group members	Wed 5/21/25	Wed 6/25/25	36	100%	26
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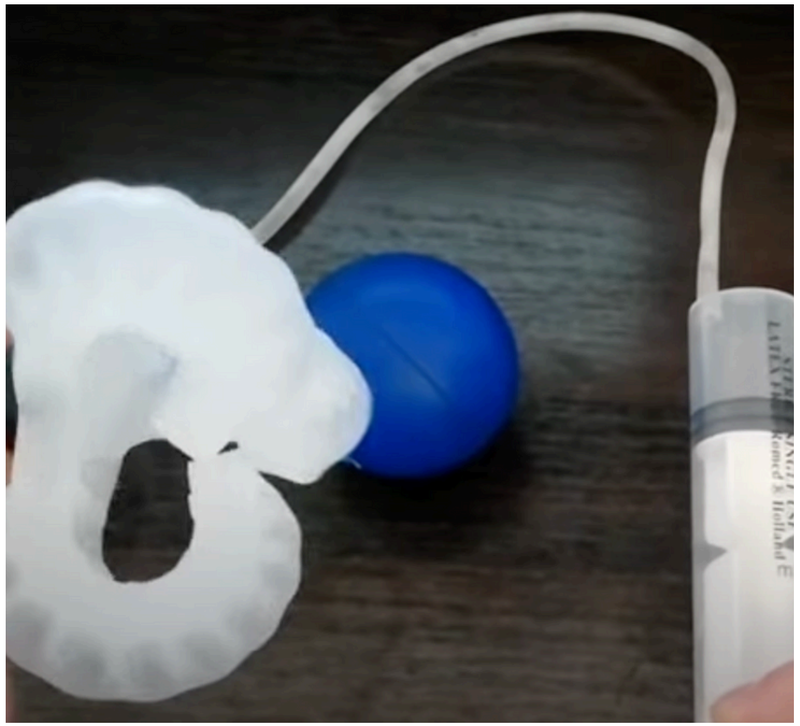
04.

Design

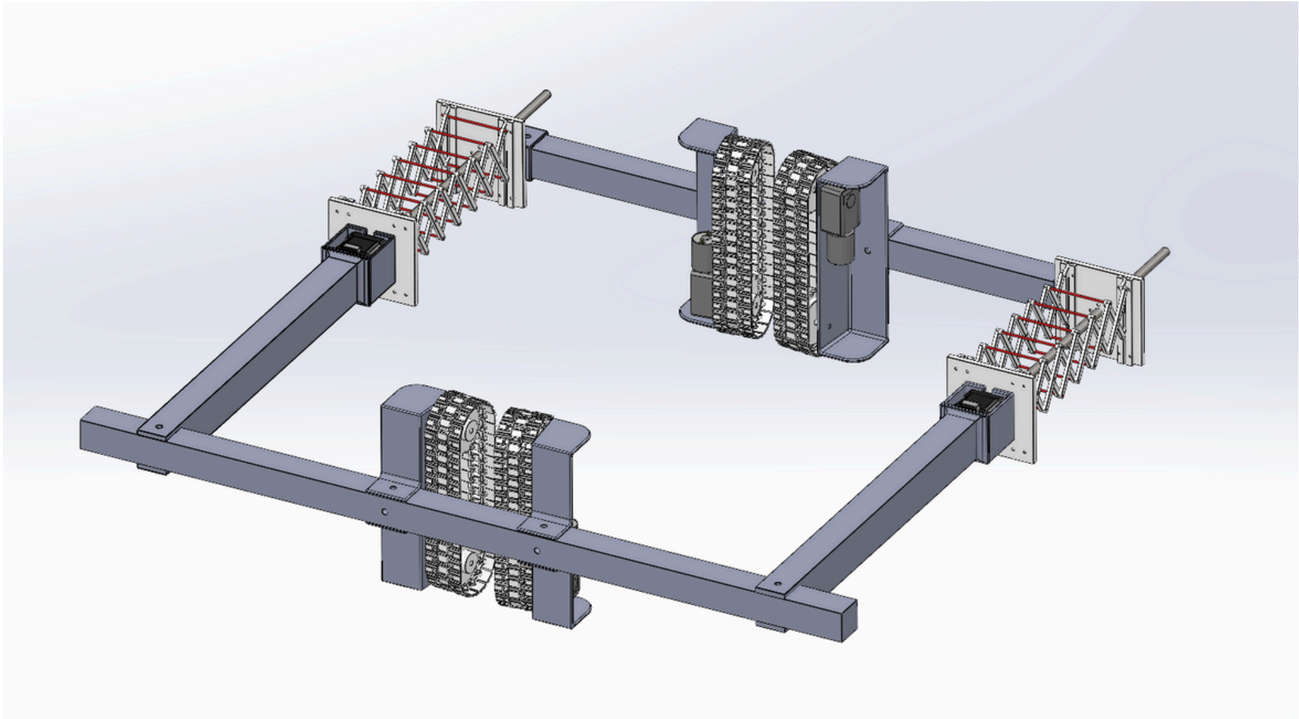


Initial Design

1

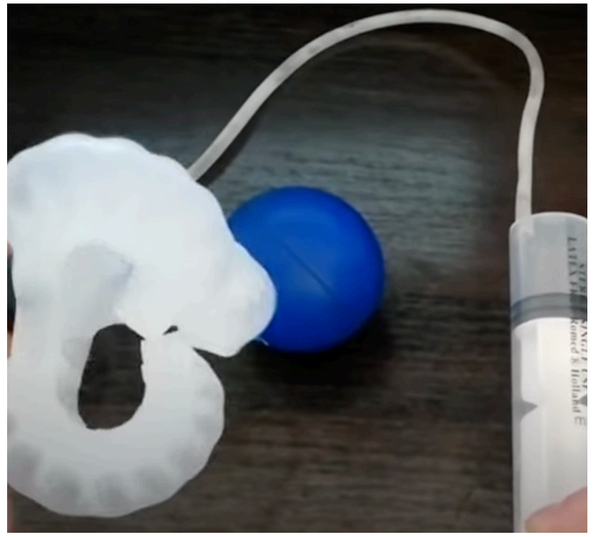


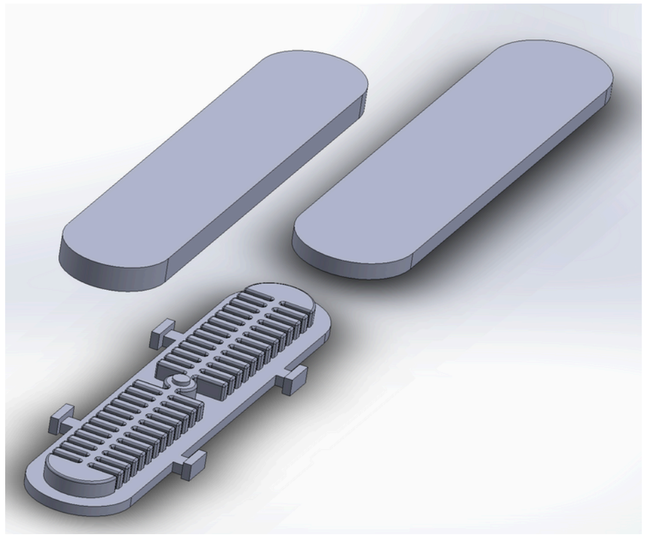
2



Initial Design 1: Soft Gripper









Design

- CAD-molded soft silicone gripper
- Air chamber bends when inflated
- Rack & pinion drives syringe for airflow

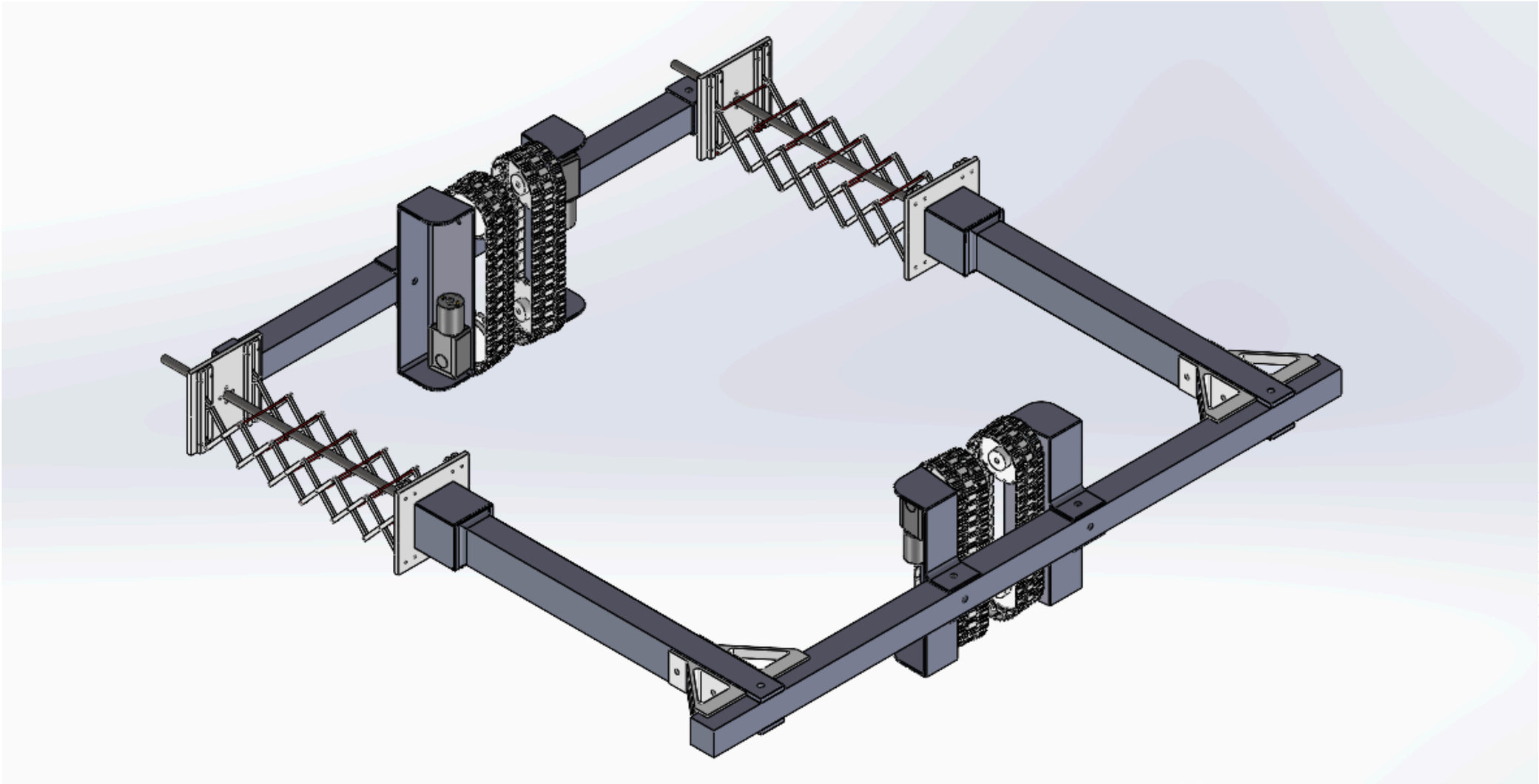
Motivation

- Soft robotics
- Better fit on curved surfaces

Why Abandoned

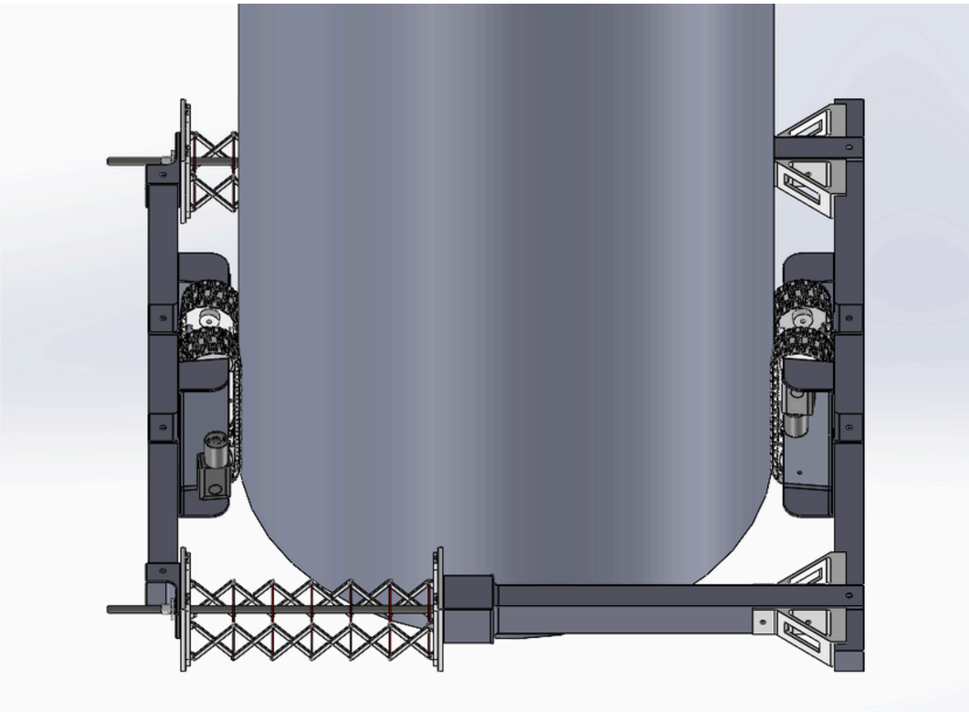
- Hard to control air volume
- Difficult to fabricate soft parts

Initial Design 2: Lead Screw-Belt Mechanism



Locking Mechanism

- 2 **lead screws** to keep the belt and the cylinder close in contact
- **Self-locking** gear motors with large torque
- **Ultrasonic sensors** to detect its diameter

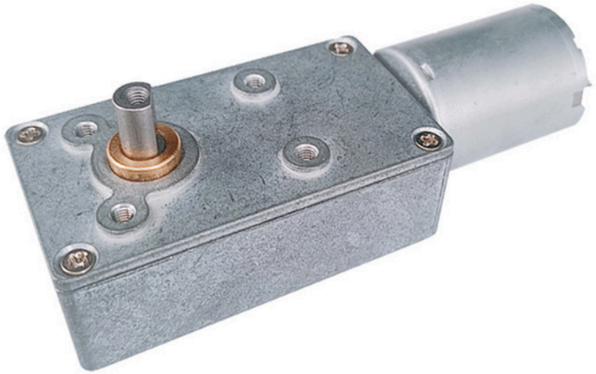


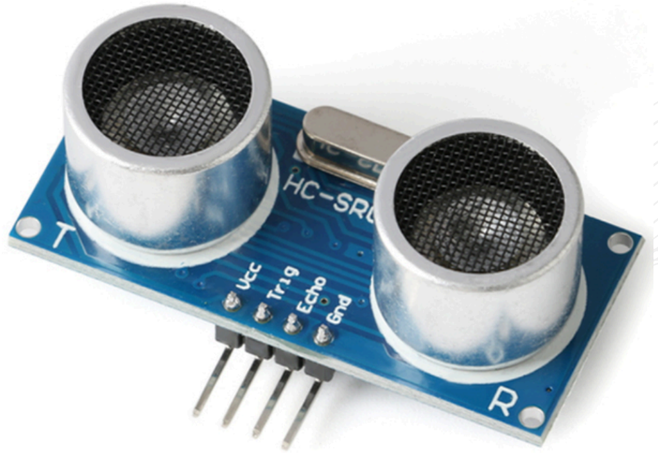


Locking Mechanism

Highlights:

- Automatic control, **no need for manual adjustment**
- Adaptable for cylinder diameters within [200 mm, 800 mm]
- Made of steel, high intensity and reliability

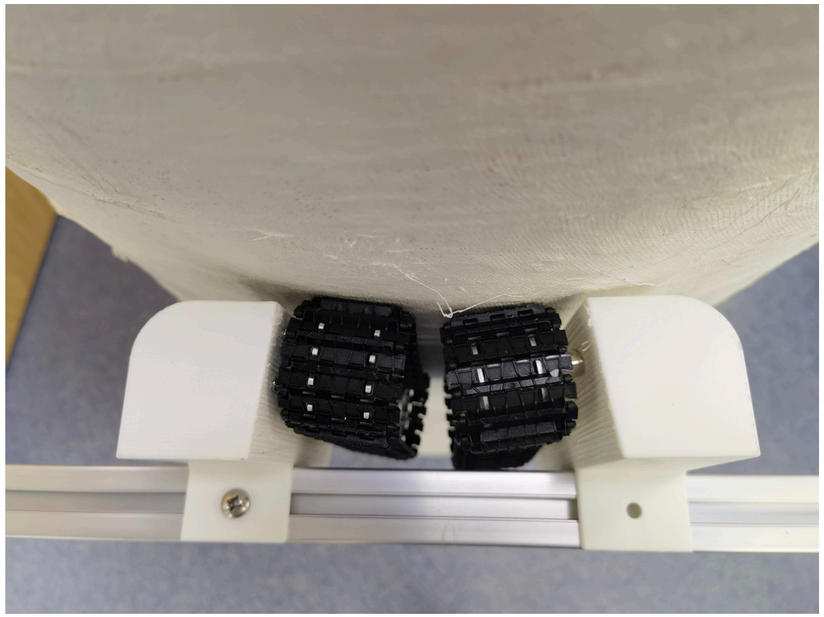


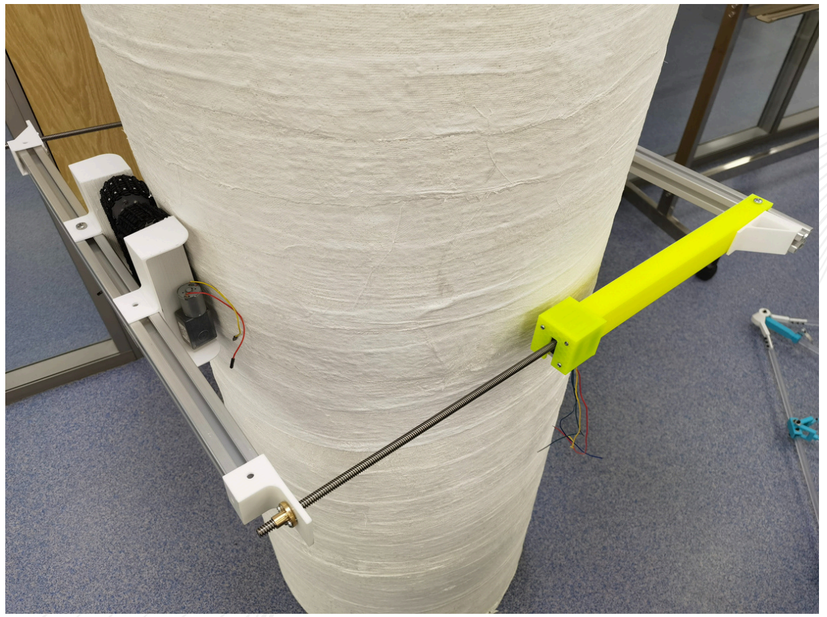


Climbing Design

- **Omni-directional wheels** at the bottom to allow for locking behavior
- 4 **belts** on the ends of the diameter of the cylinder controlled by 4 motors
- Height monitored by **laser sensor**







Adaptable Feature

- **Power-off**

The self-locking motors keep the lead screws and thus the structure at their initial position when the power is out.

- **Different width**

The lead screws have the length of 200-800mm, adjusting to different width of the cylinder.

- **Shape**

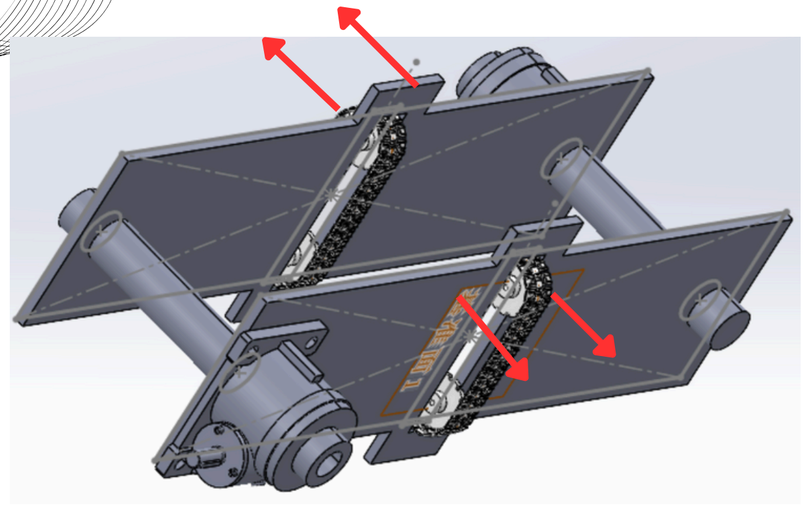
Two pairs of belts clamp the diameter of the cylinder and provide enough contact area. It can also be used on cuboid piers.

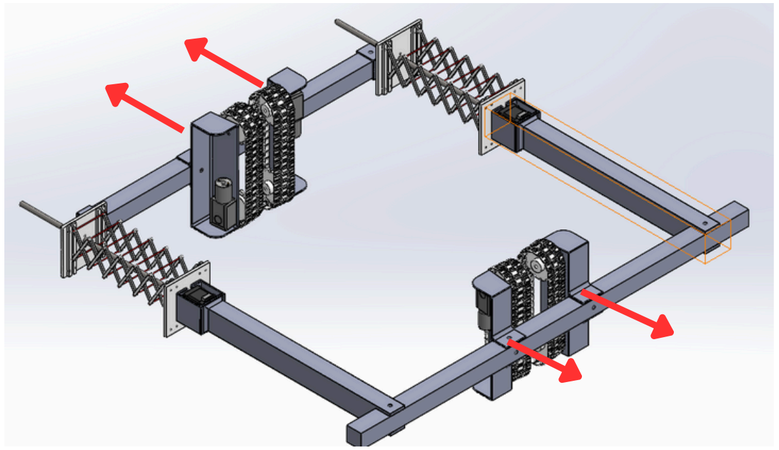
05.

Manufacture



Material selection

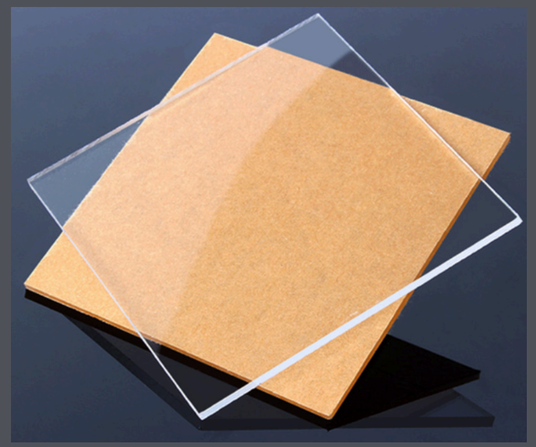




High resistance to Deflection under lateral forces

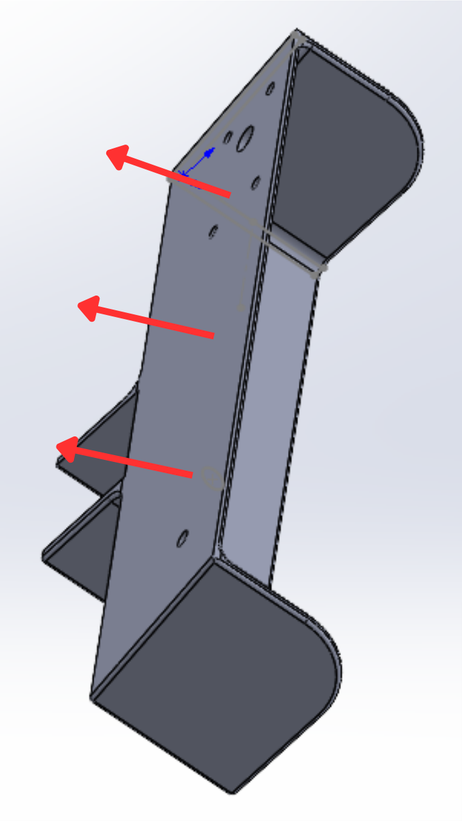
Acrylic

Aluminum Extrusions



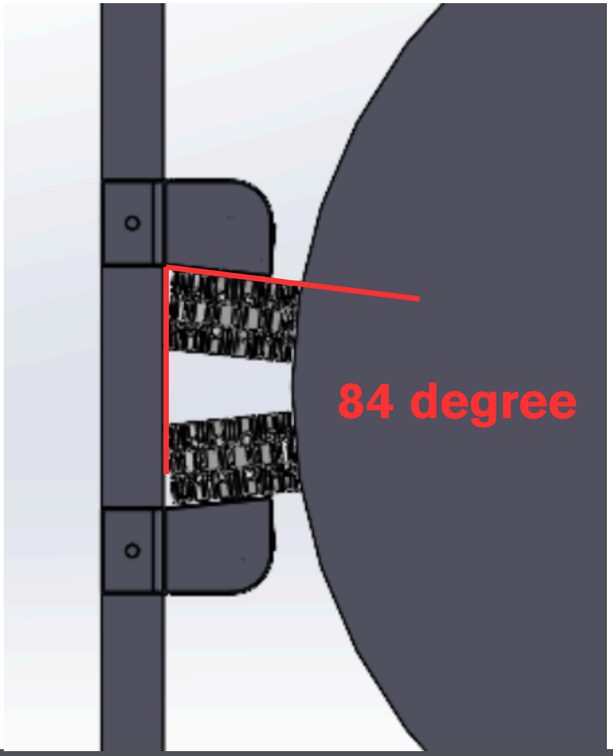


Material selection

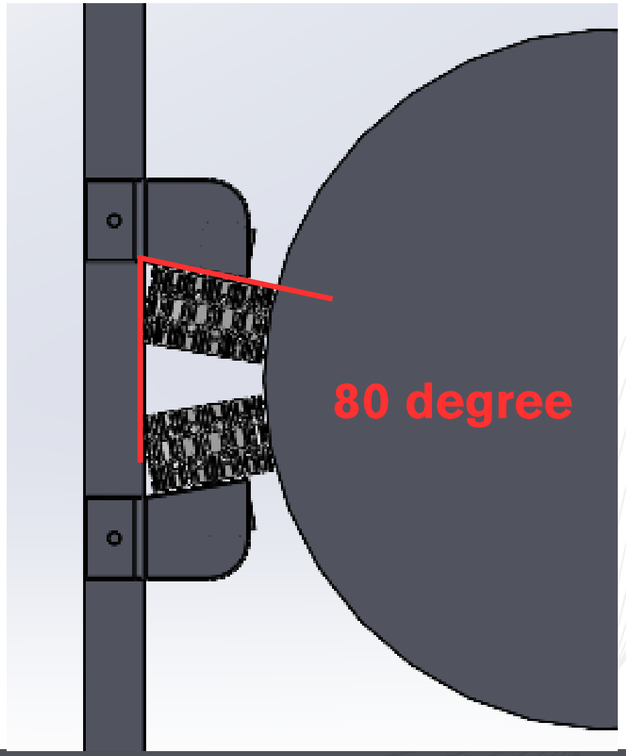


PLA

- **Resilience**
- **Able to bend**

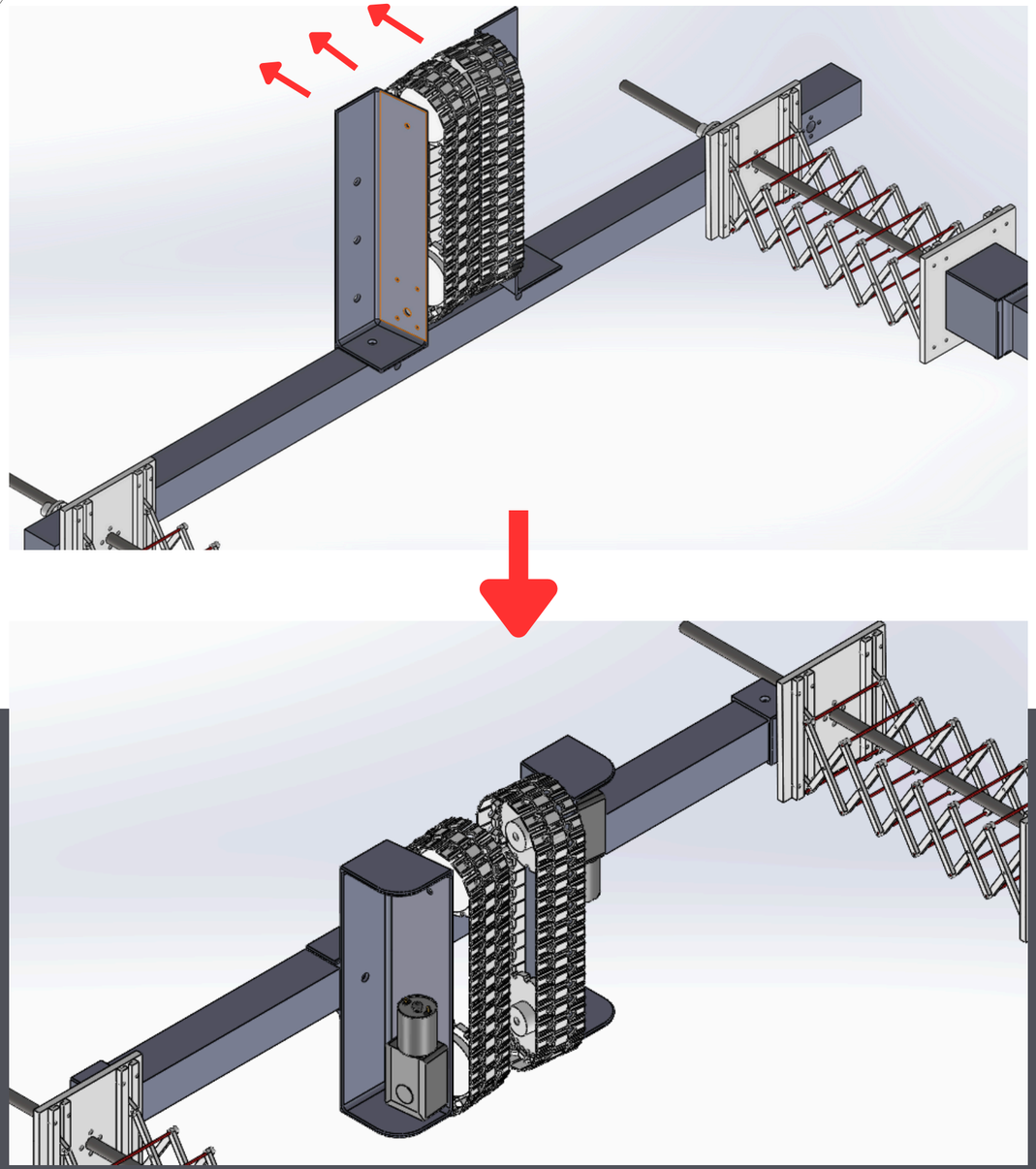


60 cm



30 cm

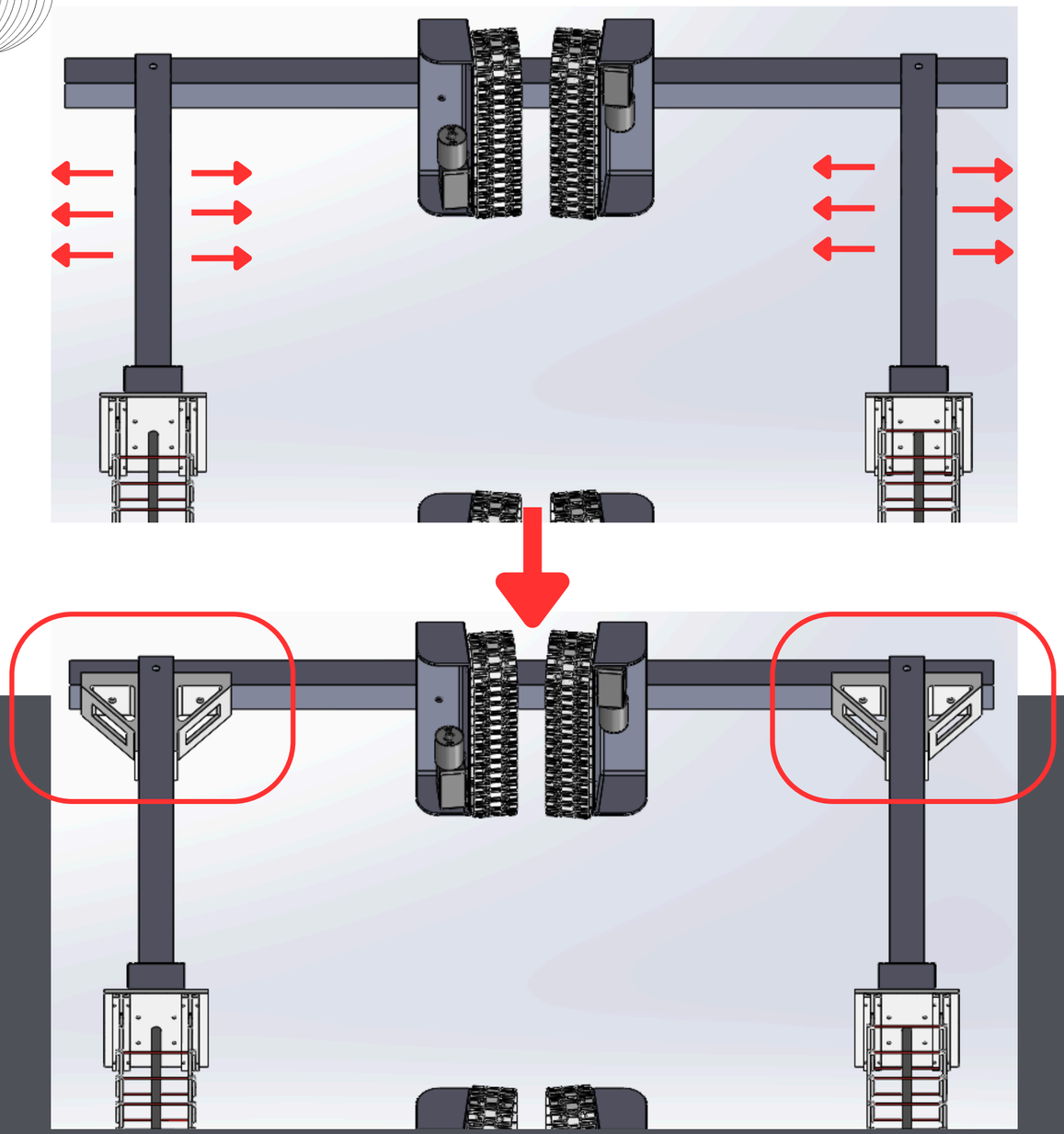
Improvements



Top-mounted Configuration

- **Uneven force distribution**
- **Bend in the direction of the arrow**

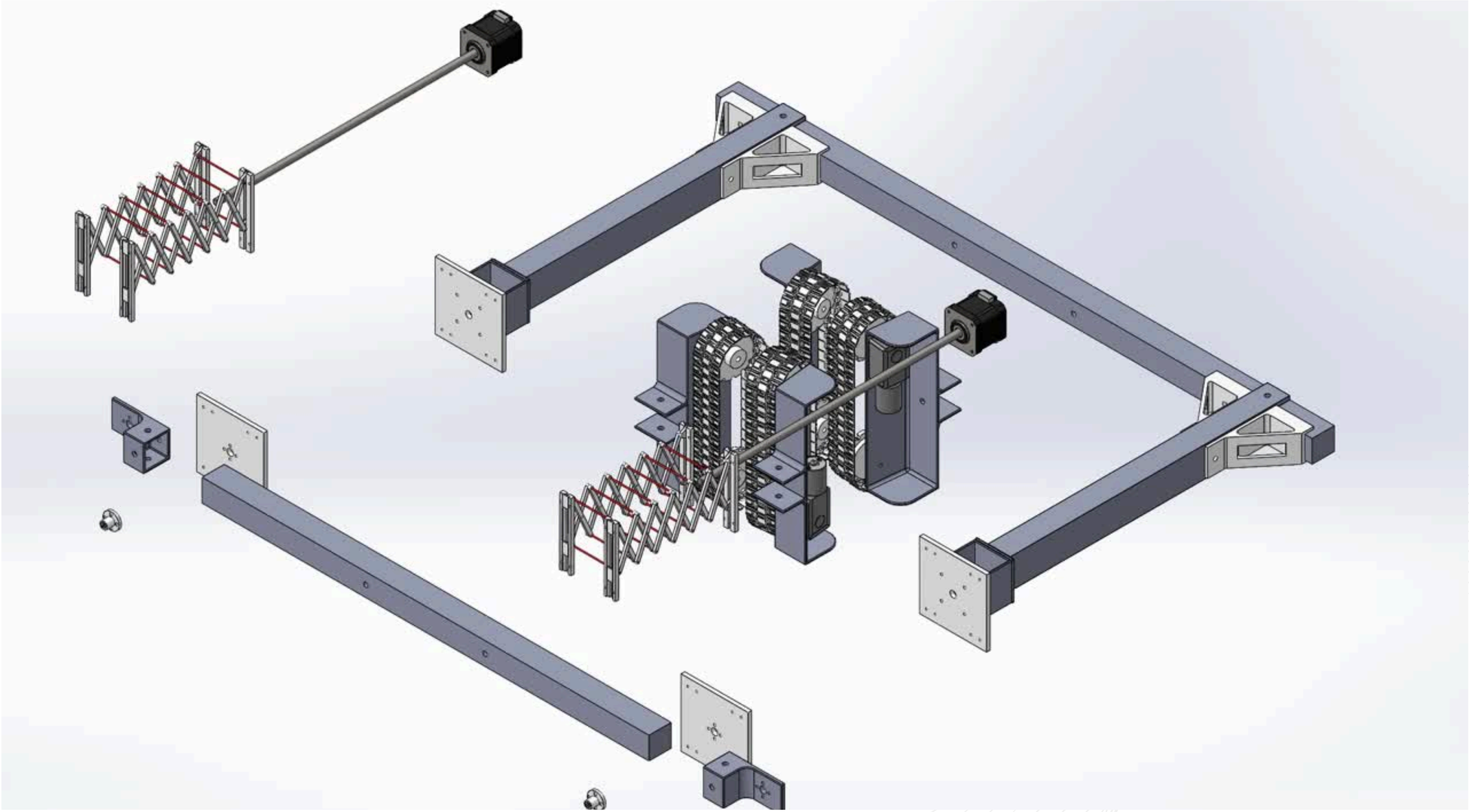
Improvements



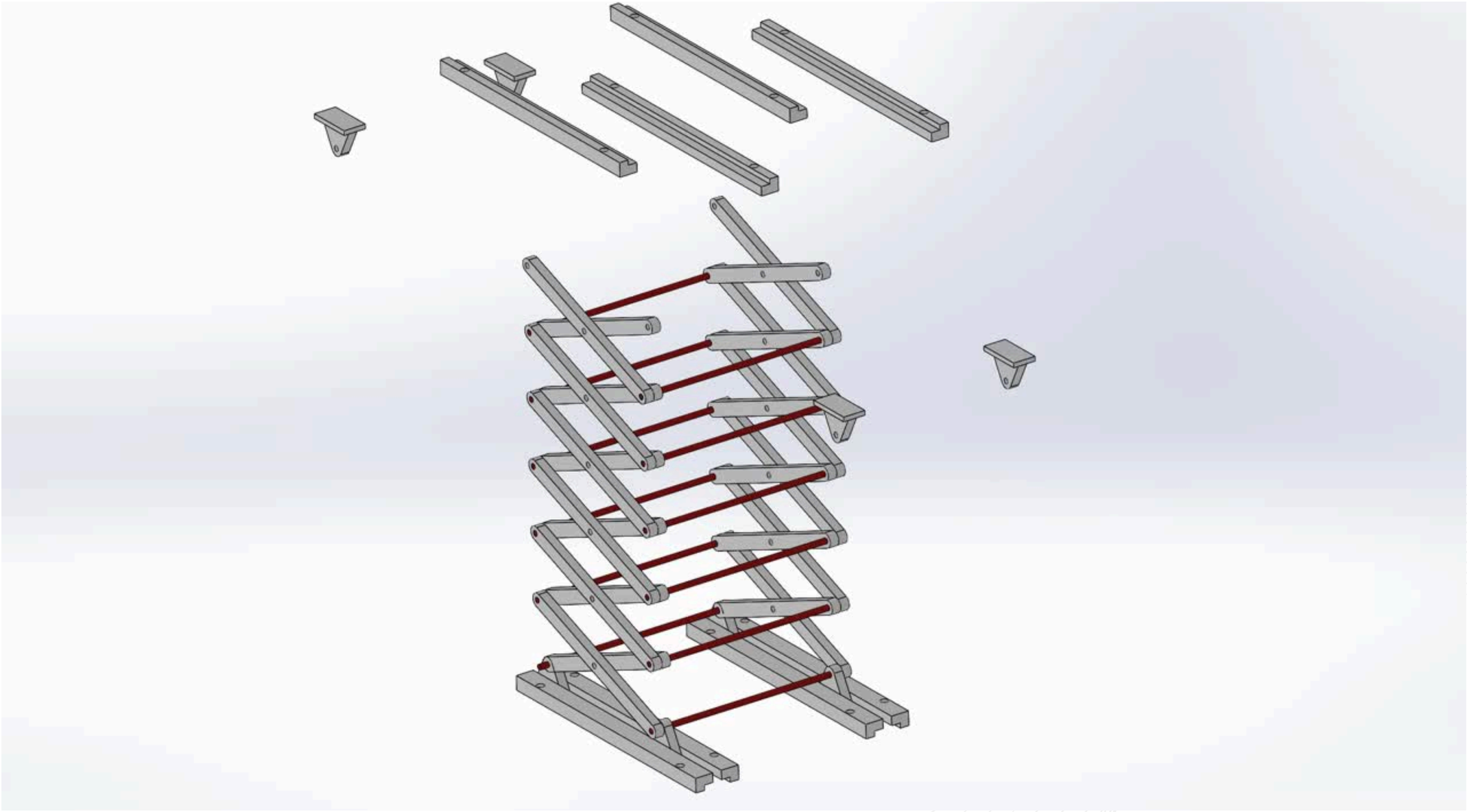
Reinforcement Parts

- **Principle of triangular stability**
- **Prevent lateral tilting**

Main Assembly

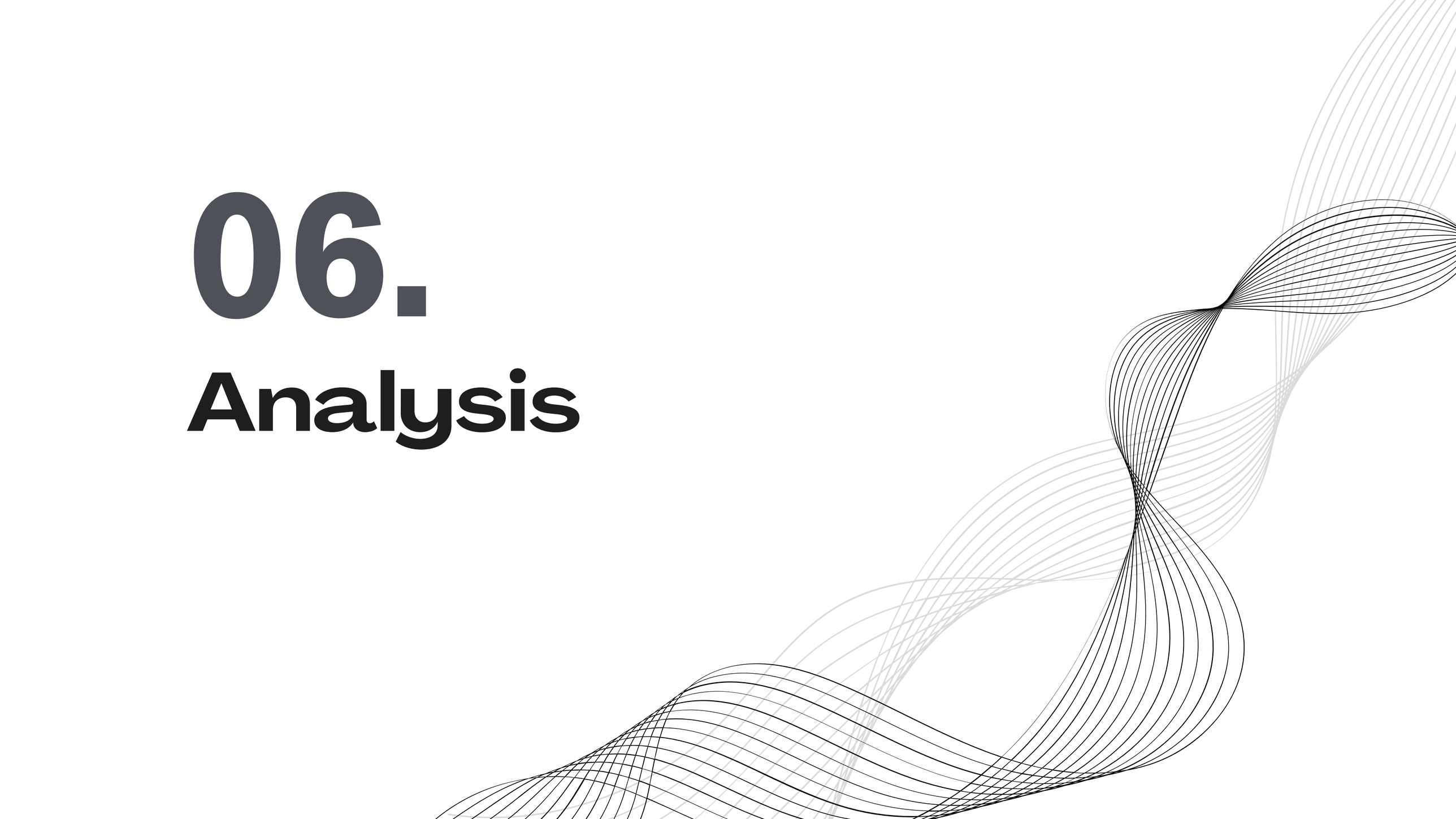


Linkage Assembly

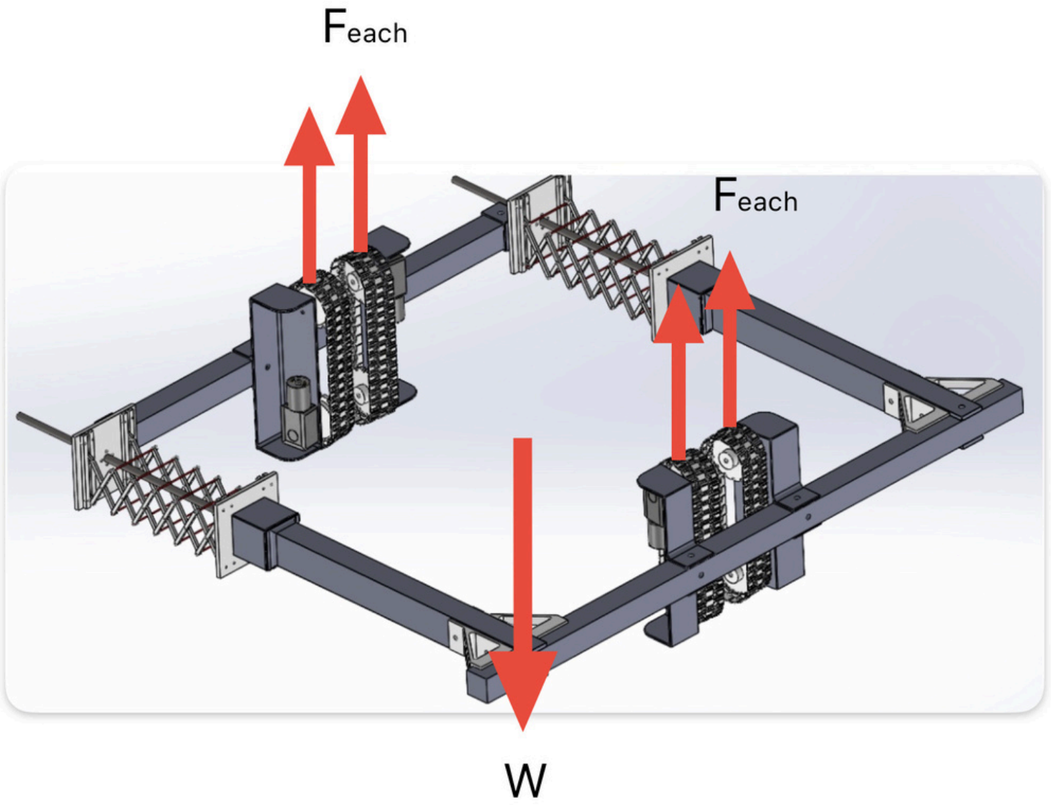


06.

Analysis



Force Analysis: Torque



The selected motor:

- 12V voltage nominal torque: $25\,\text{kgf}\cdot\text{cm} = 2.45\,\text{Nm}$
- Self-locking maximum torque: $45\,\text{kgf}\cdot\text{cm} = 4.41\,\text{Nm}$

Given parameters:

$$m = 6.2\,\text{kg}$$
$$g = 9.8\,\text{m/s}^2$$
$$W = m \times g = 6.2 \times 9.8 = 60.76\,\text{N}$$
$$n = 4$$
$$r = 2.5\,\text{cm} = 0.025\,\text{m}$$
$$\text{Safety Factor} = 2$$

Each track supports a force:

$$F_{\text{each}} = \frac{W}{n} = \frac{60.76}{4} = 15.19\,\text{N}$$

Torque required per motor (without safety factor):

$$T_{\text{each}} = F_{\text{each}} \times r = 15.19 \times 0.025 = 0.38\,\text{Nm}$$

Applying the safety factor:

$$T_{\text{required}} = T_{\text{each}} \times 2 = 0.38 \times 2 = 0.76\,\text{Nm}$$

Thus, each motor must provide at least:

$$\boxed{T = 0.76\,\text{Nm}}$$

Force Analysis: Bending

Given:

$$f = F_{\text{each}} = 15.19\,\text{N}$$

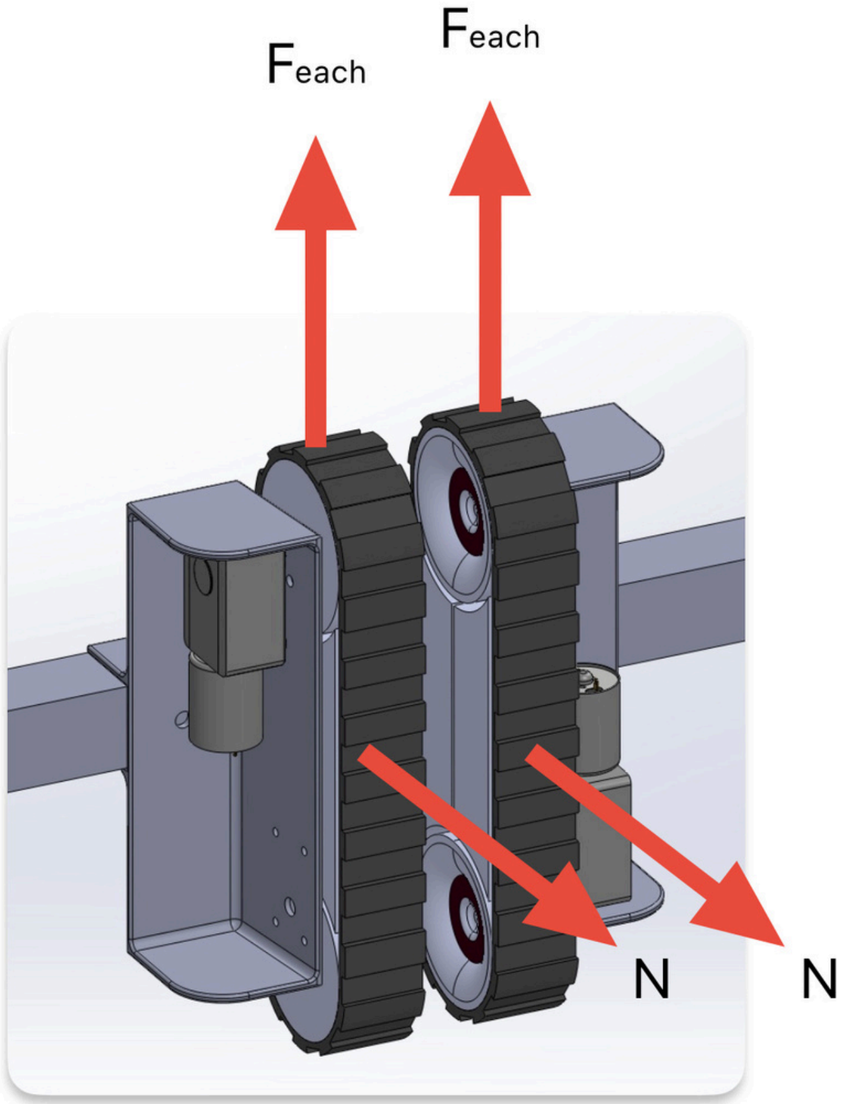
$$\mu = 0.35$$

The required normal force per track is:

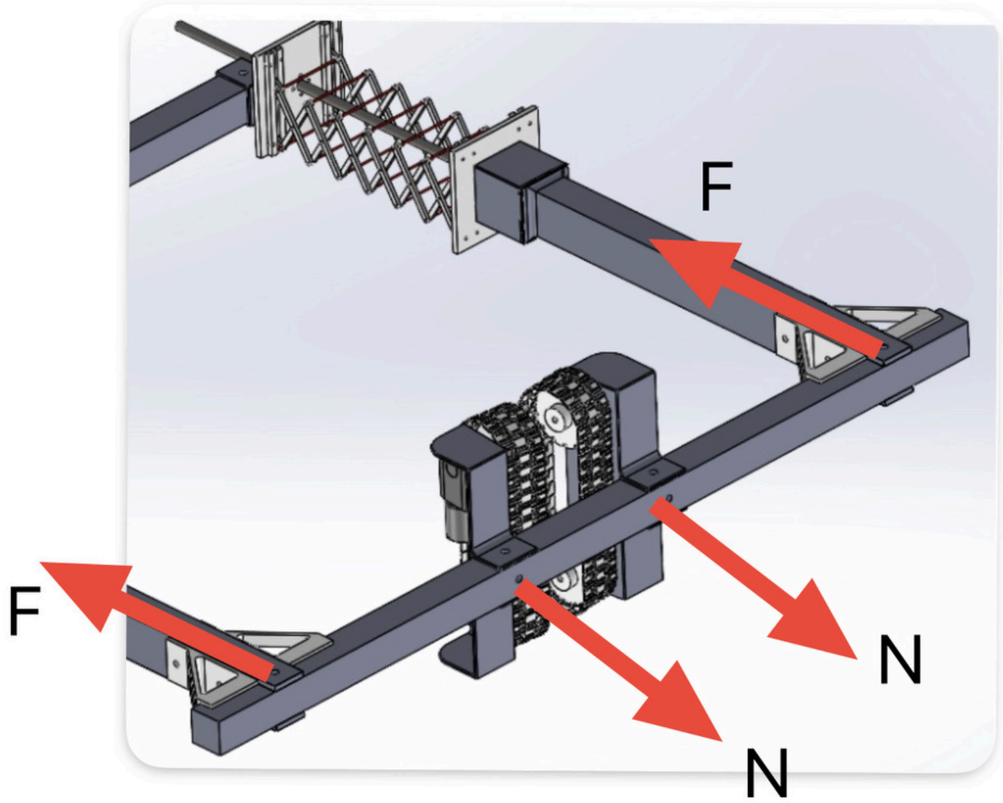
$$N = \frac{f}{\mu} = \frac{15.19}{0.35} = 43.4\,\text{N}$$

Thus, each track must apply a normal force of:

$$\boxed{N = 43.4\,\text{N}}$$



Force Analysis: Bending



Given:

$$\begin{aligned}
\text{Outer diameter:} &\quad D = 4\,\text{cm} = 0.04\,\text{m} \\
\text{Wall thickness:} &\quad t = 3\,\text{mm} = 0.003\,\text{m} \\
\text{Inner diameter:} &\quad d = D - 2t = 0.04 - 2 \times 0.003 = 0.034\,\text{m} \\
\text{Length of the beam:} &\quad L = 0.8\,\text{m} \\
\text{Applied force at each end:} &\quad N = 43.4\,\text{N}
\end{aligned}$$

Maximum bending moment:

$$M_{\max} = N \times \frac{L}{2} = 43.4 \times 0.4 = 17.36\,\text{Nm}$$

Moment of inertia for a hollow circular cross-section:

$$I = \frac{\pi}{64}\left(D^4 - d^4\right) = \frac{3.1416}{64}\left(0.04^4 - 0.034^4\right) = 6.01 \times 10^{-8}\,\text{m}^4$$

Distance from the neutral axis to the outer fiber:

$$c = \frac{D}{2} = 0.02\,\text{m}$$

Maximum bending stress:

$$\sigma_{\max} = \frac{M_{\max} \times c}{I} = \frac{17.36 \times 0.02}{6.01 \times 10^{-8}} = 5.78 \times 10^6\,\text{Pa} = 5.78\,\text{MPa}$$

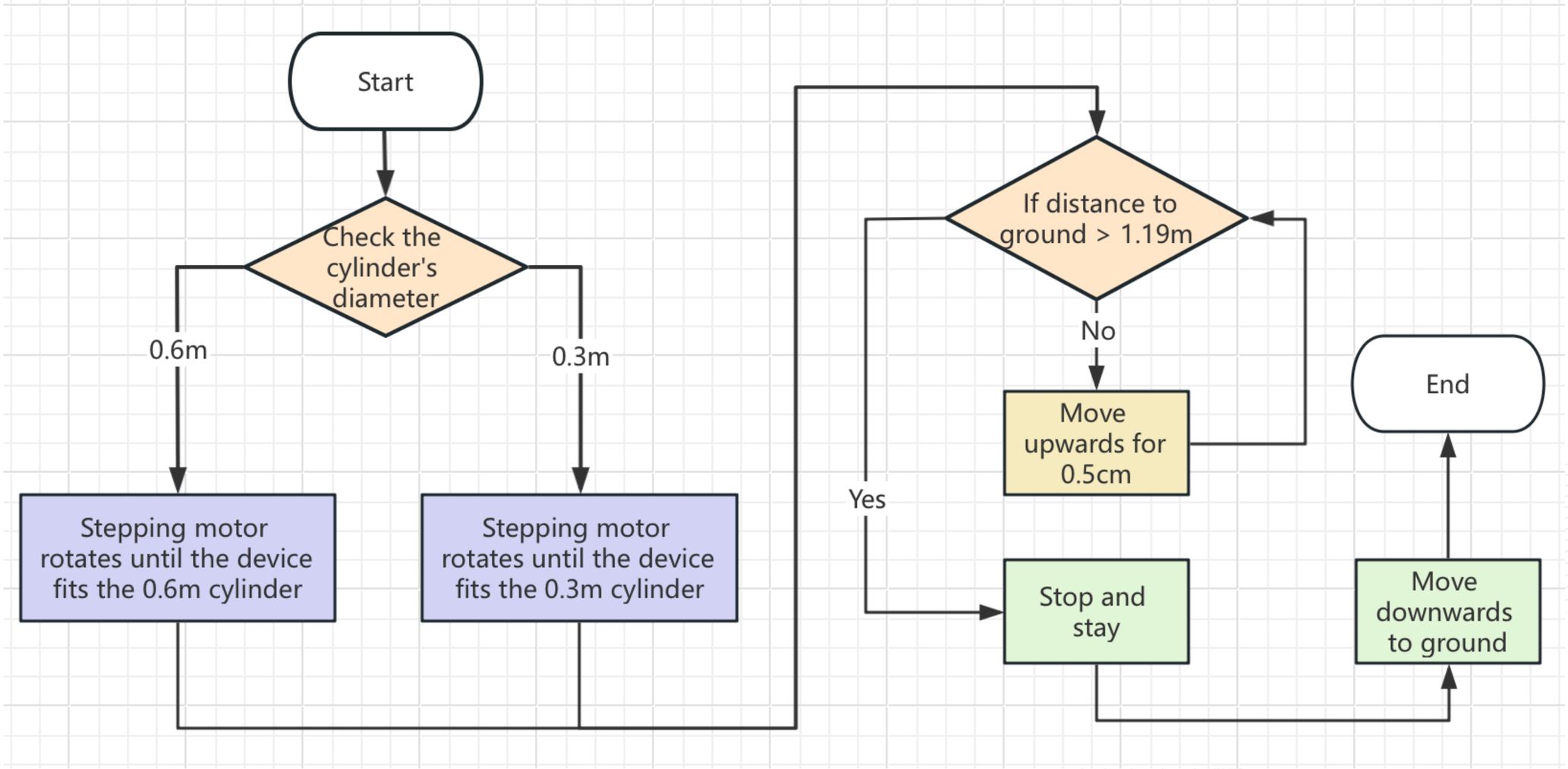
Assuming the use of 6061-T6 aluminum (yield strength approximately 276 MPa), the safety factor is:

$$\text{Safety Factor} = \frac{276}{5.78} \approx 47.8$$

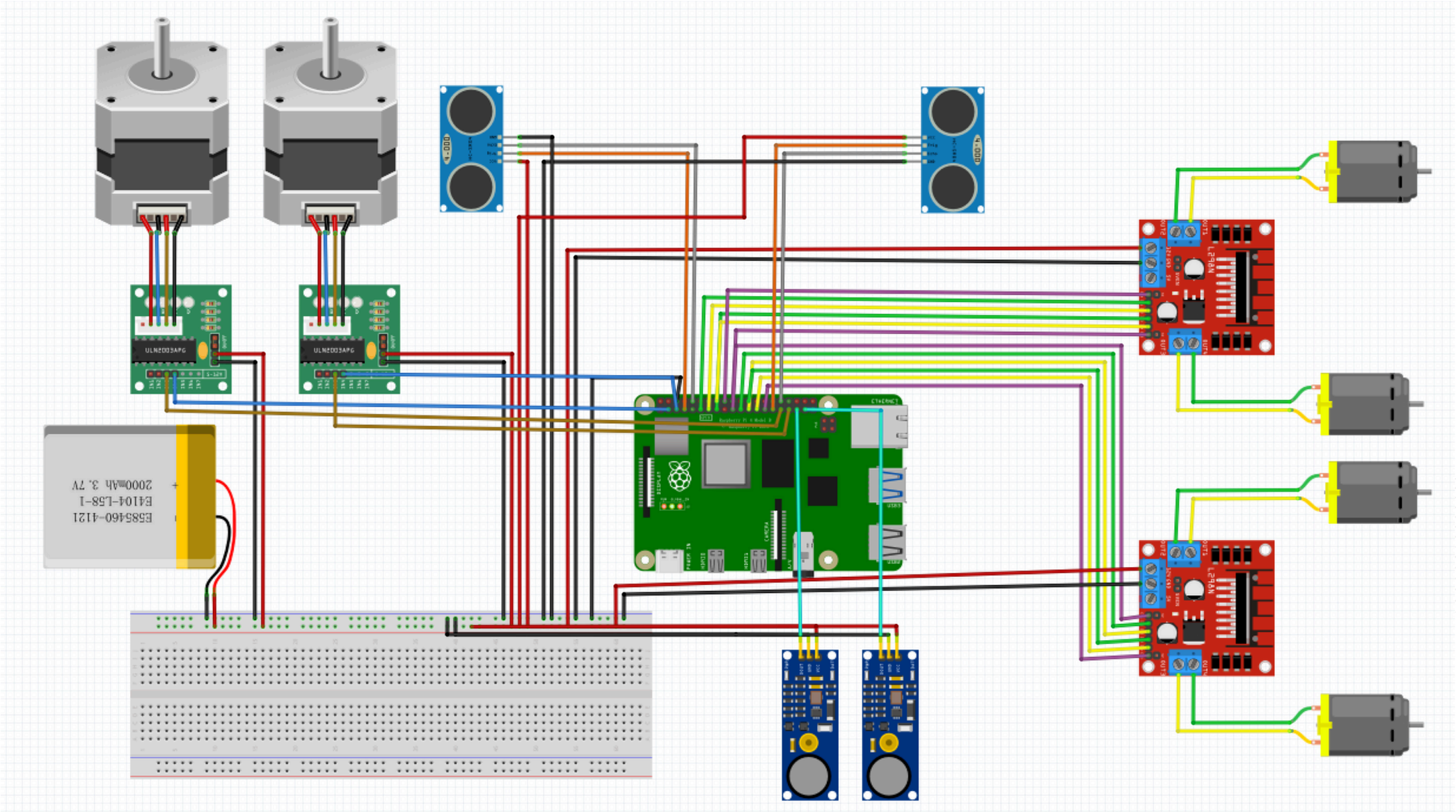
Thus, the design is very safe under the given loading conditions.

07.
Mechatronics

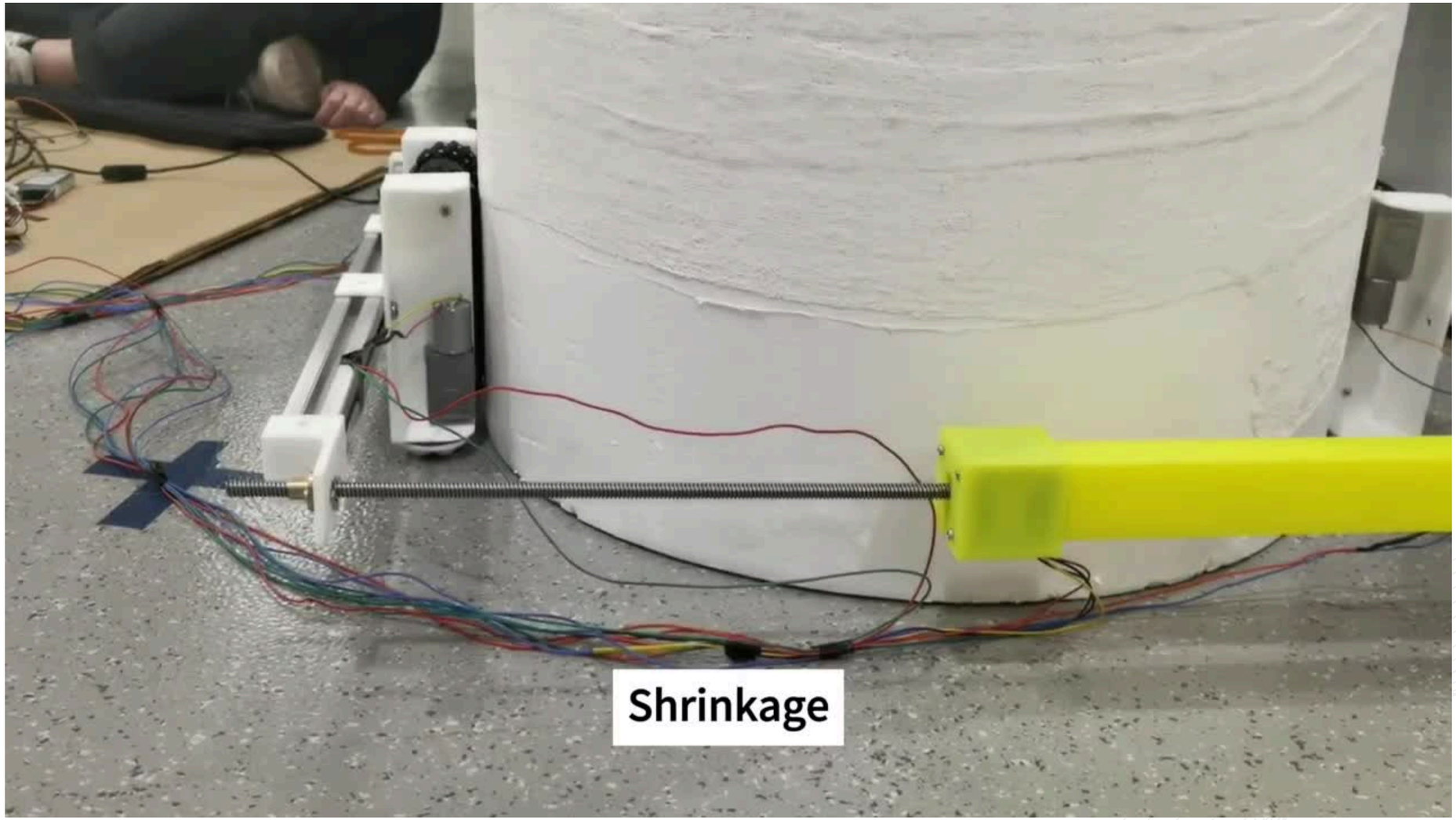
Flowchart


Start
Check the cylinder's diameter
0.6m
0.3m
Stepping motor rotates until the device fits the 0.6m cylinder
Stepping motor rotates until the device fits the 0.3m cylinder
If distance to ground > 1.19m
No
Move upwards for 0.5cm
Yes
Stop and stay
Move downwards to ground
End

Circuit Diagram



Initial Prototype



Reference

[1] G. La Rosa, M. Messina, G. Muscato, and R. Sinatra, "A low-cost lightweight climbing robot for the inspection of vertical surfaces," Mechatronics, vol. 12, no. 1, pp. 71-96, Feb. 2002, doi: 10.1016/S0957-4158(00)00046-5.

[2] J. Xue, J. Chen, A. Stancu, X. Wang, and J. Li, "Spatial Trajectory Tracking of Wall-Climbing Robot on Cylindrical Tank Surface Using Backstepping Sliding-Mode Control," Micromachines, vol. 14, no. 3, p. 548, Feb. 2023, doi: 10.3390/mi14030548.

THANK YOU

Q&A

